

Projects

1. Tool counting and Tool measurement

Client: Oerlikon Balzers

Deployed locations: 2

Use cases

1. Tool counting

- Developed a tool counting system for counting tools on a metallic tray.
- The tools would range from 10 - 600 achieving 99.98% accuracy.
- The Model is trained using only 249 images with a custom Augmentation script.
- Techniques used: Yolo Objects Detection, Augmentation using Albumentations library, Raspberry Pi, Pi camera 4MP.

2. Tool Measurement

- Developed a Tool measurement system for measuring length and diameter of the Tool.
- The tool diameter ranges from 3mm to 30mm and length ranges from 50mm to 300mm.
- The error is less than 0.5 mm for diameter and less than 1mm for length.
- Techniques used: Color segmentation, Angle Estimation using PCA, Multi Variable Linear Regression, Binary Search Algorithm, Raspberry Pi.

This project decreased the manual work and increased the productivity by 90% of the operational team.

2. Activity Assessment system

Client: Maruti Suzuki

Deployed locations: 1

Developed a computer vision-based system to assess trainer activities during assembly tasks. The system automatically monitors task execution, measures activity duration, detects idle time between steps, and identifies the transition from one activity to the next with 99.7% accuracy.

The solution helps ensure process compliance, improves training quality, and reduces manual supervision.

Activities / Use Cases Assessed:

1. Putting 4 black boxes on table - used contour detection and color thresholding
2. Splitting cards - Used motion detection and contour detection.
3. Picking up 4 screws - color segmentation and contour detection.
4. Tightening of bolts with gun - used color segmentation, sequence logic
5. Torquing of bolts with limit wrench - object detection and color segmentation and sequence logic.
6. Tightening of screw with gun - color segmentation, dilation and erosion and contour detection.
7. Fixing and defixing of hose and clamp - object detection, sequence logic, color segmentation.
8. Coupler Fitment & check - color segmentation and contour detection and image thresholding.
9. Gourmet fitment & check - color segmentation and image thresholding.
10. Fitment of earthing wire - color segmentation, contour detection.

3. Retail Store Production Monitoring

Client: Trent Limited

Deployed locations: 1200

Developed computer vision–based solutions for real-time retail monitoring and alerts. The system provides actionable insights to improve store operations, customer experience, and safety through automated detection and analysis.

Retail Monitoring Use Cases

1. Footfall count. Male female and child -- yolo object detection and object tracking.
2. Customer Dwell time. -- Reidentification of person entry and exit using resnet, embedding comparison using cosine distance
3. Fire exit blockage alert. -- used multimodal llm qwen and prompt engineering.
4. Checkout Counter Monitoring, check if all tills are active during crowd or else raise an alert.
-- Yolo object detection,
5. Backoffice cleanliness. -- used multimodal llm qwen and prompt engineering.
6. Hotspot monitoring alert. -- used multimodal llm qwen and prompt engineering.
7. Detection of gate opening and closing time -- Color thresholding
8. Store Manager Tracking using facial recognition. -- extraction of facial embeddings using resnet and keypoint matching

- 9. Intrusion Alert. -- Motion detection.
- 10. Trails overcrowd alert. -- Yolo detection

4. Star Bazaar Production Monitoring

Client: Trent Limited

Deployed locations: 3

Developed computer vision–based solutions to monitor store operations and customer movement in real time. The system helps improve store efficiency, safety, and inventory visibility by automatically detecting key activities and raising alerts

Use Cases

- 1. Footfall count. Male , female and child. -- yolo object detection and object tracking.
- 1. Fruits and Vegetables , checking for empty boxes. -- used Binary classification model using onnx.
- 2. Staples, checking for empty tins. -- used Binary classification model using onnx.
- 3. Fireexit blockage alert. -- used multimodal llm qwen and prompt engineering.
- 4. Cash till Monitoring. -- yolo object detection.
- 5. Front Gate Activity. -- Color thresholding
- 6. Store Manager Tracking using facial recognition. -- extraction of facial embeddings using resnet and keypoint matching

5. Warehouse & Dock Monitoring System

Client: Mahindra Logistics

Deployed locations: 3

Developed a computer vision–based monitoring system to track dock operations, vehicle movement, and safety compliance in logistics hubs. The solution monitors production activity, vehicle entry and exit, idle and loitering time, pallet movement, and unsafe activities near trucks.

Use Cases

1. Dock Production Monitoring (Activity, Empty , Idle time.) -- Yolo object detection
2. Truck Entry and Exit time with detection of number plate. -- Yolo object detection and tracking. Used Gemini multimodal llm to detect the number plate.
3. Suspicious Activity near the truck. -- mobilenet object detection.
4. Calculation of Loitering time. -- Used mobilenet object detection and clustering algorithm.
5. Pallet Counting. -- used mobilenet object detection and object tracking
6. Usage of mobile alert at working area. -- Used Gemini multimodal llm to detect the number plate.
7. detection of Chair or any objects at pallet moving area. -- Used yolo world object detection which uses clip model.
8. Deployed in Pune, Jaipur, Hyderabad.

6. Patient & Operations Monitoring System

Client: Dr Batras

Deployed locations: 11

Developed a computer vision–based monitoring system for clinics to analyze patient flow, staff behavior, and service quality. The solution tracks customer visits, waiting and consultation time, staff compliance, greetings, cleanliness, and device health, helping improve patient experience and clinic operations through real-time insights and alerts.

Use Cases

1. Male and Female footfall count. -- Yolo object detection using ncnn.
2. Detection of unique customers entered inside the clinic using person reidentification. -- Reidentification of person entry using resent, embedding comparison using cosine distance
2. Customer Dwell Time. -- Reidentification of person entry and exit using resent, embedding comparison using cosine distance
3. Time spent by customers with the doctor. -- Reidentification of person entry and exit using resent, embedding comparison using cosine distance
4. Detection of staff without the uniform. -- color thresholding and Reidentification
5. Check if the customer is greeted. -- Yolo object detection.
6. Audio analysis of customer interaction with Doctor. -- used Multimodal gemini and prompt engineering.
7. Check how long the receptionist is present during working hours to check with patients. -- yolo object detection.
8. Check if the store is neat and clean. -- used Multimodal gemini and prompt engineering.

9. Complete live status of clinics cameras and devices. raise an alert in case of offline status to avoid data loss. -- used devices metadata.
10. Detect store opening and closing time. -- grayscale detection.

7. Warehouse Production Monitoring

Client: Ultratech Cement

Deployed locations: 2

Developed a computer vision–based production monitoring system to track the movement of cement bags inside and outside the warehouse. The solution automatically counts bags during entry and exit, helping improve inventory visibility and reduce manual tracking.

Use Cases

1. Counting cement bags entering the warehouse – Mobilenet Object detection and tracking.
2. Counting cement bags exiting the warehouse – Mobilenet Object detection and tracking.

8. Customer Analytics System

Client: Royaloak

Deployed locations: 86

Developed a computer vision–based customer analytics solution to accurately count customer groups and identify unique visitors entering the store. The system avoids duplicate counting by re-identifying customers across frames, providing reliable footfall insights.

Use Cases

1. Group counting of customers entering the store -- used mobilenet object detection, object tracking
2. Unique customer counting using re-identification – Reidentification of person entry to remove duplicate counting with resnet, embedding comparison using cosine distance

